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ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)

ORGANISATION OF ISLAMIC COOPERATION (OIC)

Department of Computer Science and Engineering (CSE)

SEMESTER FINAL EXAMINATION SUMMER SEMESTER, 2023-2024

DURATION: 3 HOURS FULL MARKS: 150

CSE 4835: Pattern Recognition

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer **all 6 (six)** questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

1. a) Justify your agreement/disagreement with the following statements: 3×5 (CO2) (PO2)

    i. During the training process, if the loss curve is observed to plateau at a relatively high value in the early stages, one possible refinement is to increase the learning rate.

    ii. One theorem says, "A Neural Network with one hidden layer can approximate any function $f : R^N \rightarrow R^M$ with arbitrary precision." The same theorem is applicable for a Convolutional Neural Network (CNN) with one hidden layer.

    iii. Assume that, in a classification problem, there are two classes: Square and Circle. Figure 1 represents the data points of the two classes. Here, each axis represents one feature. The position of the data point along that axis represent the magnitude of the feature for that particular datapoint. The training set is represented by filled (grey) markers, while the test set is represented by unfilled markers. In this scenario, a linear SVM can achieve 100% accuracy on both the training and test sets.

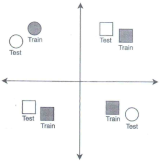

Figure 1: Sample datapoints of two classes for Question 1.a (iii)

    iv. After applying max pooling with spatial extent 2, stride 2, and zero padding on an input with $W \times H \times C$, (where W, C, and H are all divisible by 2), the number of values in the output shape is one eighth that of the input.

    v. ReLU activation is better than Sigmoid because it is zero-centred.

   b) Consider the following keywords: {Score, Weight vector, Gradient Descent, Loss Function, Input data, Backpropagation, Regularization} $5 + 5$ (CO1) (PO1)

    Draw a flowchart by arranging the aforementioned keywords according to their roles in solving a classification problem. Discuss their roles and relations with each other in brief.

2. a) Figure 2.A shows a loss curve produced by a 'black box optimizer'. Figure 2.B and 2.C show the state of the loss curve after tuning a hyperparameter. 4×3 (CO2) (PO2)

   i. Which hyperparameter is likely to be modified here?
   ii. Out of the Experiments A, B, and C, which one corresponds to the largest magnitude of the hypermeter? Justify.
   iii. The loss curve for Experiment C seems to be the most desirable. Despite this, is there any reason for which someone would choose the hyperparameter in Experiment B for training a model?

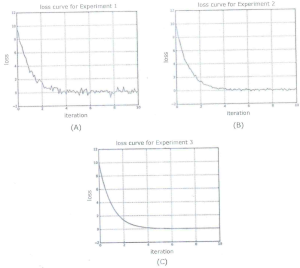

Figure 2: Loss curves of Experiments A, B, and C for Question 2.a

b) You have a linear classifier with weights W as follows: $4 + 4$ (CO2) (PO2)

$$W = \begin{bmatrix} 3 & -1 & -2 \\ -3 & 1 & 4 \end{bmatrix}$$

This classifier has been trained to classify data points of the form $x = [a, b, c]$, where a represents how spiky a shape is, b represents how large a shape is, and c represents how many dots the shape has.

The classifier will classify the shape as either a 'Wob' (class 1) or a 'Bob' (class 2). So the classifier works as follows:

$$Wx = \begin{bmatrix} \text{'Wob' Score} \\ \text{'Bob' Score} \end{bmatrix}$$

i. Classify the data point $x = [1, 3, -1]$.

ii. Assume that the given weights (W) can correctly capture the features of the true objects. Based on the weights, determine class labels of the two shapes given in Figure 3 with appropriate justification.

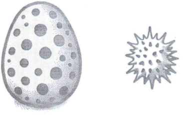

Figure 3: Data Sample for Question 2.b

c) Model ensembling usually involves training multiple independent models and then providing a final decision based on probability averaging or a max-voting strategy. Suppose you cannot afford the computational resource to train multiple models. How can a cyclic learning rate scheduler be utilized in this situation to avoid multiple model training? 5 (CO1) (PO1)

3. a) Explain the working procedure of the Fast R-CNN model in detail. 10 (CO1) (PO1)

b) A retail company wants to develop a pattern recognition system to automatically classify products based on images taken in their warehouses. Initially, the team trains a Convolutional Neural Network (CNN) from scratch using 20,000 labeled product images. However, the model shows poor generalization when tested on new product categories and images taken under different lighting conditions. Collecting and annotating a larger dataset is costly and time-consuming. 5×3 (CO2) (PO2)

One of the senior engineers suggests applying transfer learning. The idea is to take a CNN pretrained on a large-scale dataset such as ImageNet and then fine-tune it on the companys product dataset. Another engineer argues that feature extraction with frozen layers may be more efficient, given the limited compute resources. Meanwhile, the project manager is concerned about the trade-off between model accuracy and training time, since the solution must be ready within a strict deadline.

Based on the given scenario, answer the following questions:

i. Explain how transfer learning could help in this scenario compared to training a model from scratch.

ii. Differentiate between fine-tuning and feature extraction with frozen layers in the context of transfer learning. Which approach would you recommend here, and why?

iii. Discuss the potential limitations or risks of applying transfer learning in this case.

4. a) Consider Figure 4 to answer the following questions: 5×3 (CO2) (PO2)

   i. Discuss how the design choices considered inside the Squeeze-and-Excitation block (SE-block) enhances the performance of a vanilla Bottleneck Residual Block.

   ii. Analyze the computational impact in terms of FLOPs count due to the inclusion of the SE-block.

   iii. Criticize the idea of replacing the Sigmoid activation in the SE-block with ReLU.

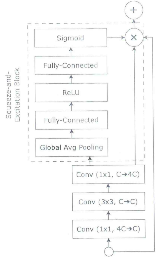

Figure 4: Bottleneck Residual Block with Squeeze and Excitation Module for Question 4.a

   b) With necessary diagrams, discuss the limitations of building Convolutional Neural Networks (CNNs) via Stacking Grouped Convolution and how ShuffleNet addresses this issue. $5 + 5$ (CO1) (PO1)

5. a) 'MobileNets run extremely fast on CPUs, but it may run slower on GPUs' - Justify the statement. 5 (CO1) (PO1)

   b) Discuss the role of the Non-Max Suppression algorithm in object detection, along with its limitations. $3 + 2$ (CO1) (PO1)

   c) Explain the detailed working mechanism of a Feature Pyramid Network (FPN) in the context of object detection. Furthermore, discuss how the underlying principles of FPN can be adapted for enhancing image classification backbones and under what circumstances such an adaptation may be beneficial. $10 + 5$ (CO1) (PO1)

6. A group of young researchers is developing a robust pipeline for plant disease classification from leaf images. The dataset contains 18,000 samples belonging to 10 classes. However, class imbalance exists, as one class contains approximately 5,000 samples, whereas another class has only 200 samples. To address this issue, they considered three strategies: upsampling, downsampling, and a hybrid approach. In the upsampling strategy, all other classes are augmented to match the highest sample count. Conversely, the downsampling strategy reduces the number of samples in the majority classes to achieve balance. The hybrid approach applies moderate downsampling to the majority classes while simultaneously employing augmentation-based upsampling for the minority class.

   After selecting one of these balancing strategies, the researchers employ a suitable backbone network and train the model for 1,000 epochs with a learning rate of 0.001. Additionally, to further enhance model performance, several dense layers are incorporated after the flattened activation map from the backbone.

   Based on the given scenario, answer the following questions:

   a) Criticize the three strategies utilized to tackle the dataset imbalance issue. 5 (CO2) (PO2)

   b) Instead of adopting one of those three strategies, propose an alternative approach. 5 (CO3) (PO3)

   c) Criticize the decision of backbone training with 1000 epochs with a learning rate of 0.001, and propose a better strategy. 5 (CO2) (PO2)

   d) Highlight the potential usefulness of the dense layers incorporated at the end of the backbone and some design guidelines to mitigate the risk of overfitting. 5 (CO3) (PO3)

   e) From the different techniques learned in this course, recommend additional strategies that could potentially improve model performance in this context. 5 (CO3) (PO3)